01. PROGRAMMING & SOFTWARE ENGINEERING

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Python

- Fundamentals
- OOP
- Async Python
- Type Hints
- Error Handling
- Testing

Backend

- REST APIs
- FastAPI
- Pydantic
- Authentication
- WebSockets
- API Design

Data

- SQL
- PostgreSQL
- Redis
- JSON
- Data Modeling

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Dev Engineering

- Git / GitHub
- Docker
- Logging
- Linux
- Environment Management
- Cloud Fundamentals

Code · Build · Learn · Repeat

02. AI / ML FOUNDATION

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Machine Learning

- Supervised ML
- Unsupervised ML
- Features
- Evaluation
- Model Serving

Deep Learning

- Neural Networks
- Backpropagation
- Optimizers
- PyTorch
- GPU Basics

NLP

- Tokenization
- Embeddings
- Attention
- NLP Basics

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Transformers

- Self Attention
- Positional Encoding
- Pretraining
- Inference
- Multi-Head Attention
- Encoder / Decoder
- Fine-Tuning

From Data to Intelligence

03. GENERATIVE AI / FOUNDATION MODELS

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LLMs

- GPT / Gemini / Claude
- Open Models
- Tokens
- Context Window
- Inference

Model Engineering

- API Usage
- Model Selection
- Quantization
- Fine-Tuning
- LoRA / PEFT

Multimodal

- Vision
- Audio
- Speech
- Documents

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Context Engineering

- Prompt Engineering
- Few-Shot Prompting
- Dynamic Context
- Context Selection
- System Prompts
- Structured Context
- Context Compression
- Long-Context Strategies

Better Context = Better Results

04. LLM APPLICATION ENGINEERING

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LLM Application Engineering

01 Structured Outputs

02 JSON Schemas

03 Function / Tool Calling

04 Streaming

05 Async Execution

06 Retry / Fallback

07 Model Routing

08 Semantic Caching

09 Rate Limiting

10 LLM Gateways

11 Cost Optimization

Build Production-Ready LLM Apps

05. RAG ENGINEERING

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Ingestion

- Parsing
- OCR
- Chunking
- Metadata
- Indexing

Retrieval

- Embeddings
- Vector Search
- Vector DB
- Metadata Filters
- Similarity Search

Generation

- Context Building
- Citations
- Grounding
- Answer Generation

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Advanced RAG

- Hybrid Search
- Query Expansion
- Multi-Query RAG
- Graph RAG
- Corrective RAG
- Agentic RAG
- Reranking
- Query Rewriting
- Parent-Child Retrieval
- Multimodal RAG
- Self-RAG

Your Data + AI = Knowledge

06. AI AGENTS

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Tools

- APIs
- Database
- Search
- Browser
- Code
- Files

Memory

- Short-Term
- Long-Term
- Episodic
- Semantic
- User Memory

Reasoning

- Planning
- Reflection
- ReAct
- Task Decomposition

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MCP — standard way for agents to use external tools and resources

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MCP Client

- Connects to MCP servers and uses tools

MCP Server

- Exposes tools, resources and prompts

Tools

- Functions that agents can use

Resource

- Data, files, APIs and external systems

Prompts

- Reusable prompts and templates

Security

- Access control, authenticatio and safe execution

Plan · Reason · Use Tools · Get Things Done

07. AGENT ORCHESTRATION

Agent State Management

- State
- Checkpoints
- Persistence
- Context

Workflows

- Sequential
- Parallel
- Conditional
- Loops
- Event Driven

Routing

- Agent Selection
- Tool Selection
- Dynamic Routing
- Model Routing

MULTI-AGENT SYSTEMS

Supervisor Agent

- Planner → Worker
- Manager → Specialists
- Agent → Agent
- Agent → Human

Handoffs

- Task Transfer
- Context Sharing
- Seamless Transitions
- Inter-Agent Handoffs

Collaboration

- Shared Goals
- Shared Memory
- Collective Problem Solving
- Team of Agents

Human-in-the-Loop

- Approval
- Review
- Escalation
- Intervention

Frameworks

- LangGraph
- OpenAI Agents
- Google ADK
- PydanticAI

From Single Agent to Multi-Agent Systems

08. AI EVALUATION ENGINEERING

LLM Eval

- **Accuracy**
  Correctness of the response
- **Relevance**
  How relevant is the answer
- **Hallucination**
  Detect false or made-up content
- **Toxicity**
  Harmful, biased or unsafe content
- **LLM-as-Judge**
  Use LLMs to evaluate responses

RAG Eval

- **Retrieval**
  Are the right documents retrieved?
- **Recall**
  Relevant documents found
- **Precision**
  Irrelevant documents filtered
- **Groundedness**
  Is the answer supported by context?
- **Faithfulness**
  Does it stay true to the source?

Agent Eval

- **Task Success**
  Did the agent achieve the goal?
- **Tool Selection**
  Did it choose the right tools?
- **Tool Arguments**
  Are the tool inputs correct?
- **Trajectory**
  Is the plan and reasoning sound?
- **Goal Completion**
  End-to-end task completion

Evaluation Pipeline — systematic way to test, compare and improve AI systems

Golden Dataset

- High quality annotated data

Offline Evals

- Evaluate before deployment

Online Evals

- Evaluate with real users

A/B Testing

- Compare different models / approaches

Regressic Testing

- Catch performance degradation

Human Evaluation

- Human review and feedback

Measure · Validate · Improve · Build Trust

09. AI OBSERVABILITY

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Traces

- Agent Trace
- LLM Calls
- Tool Calls
- Retrieval
- Handoffs

Metrics

- Latency
- Tokens
- Cost
- Success Rate
- Failure Rate

Logs

- Errors
- Requests
- Tool Logs
- Audit Logs

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OpenTelemetry

- Traces
- Metrics
- Spans
- Semantic Conventions

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Platforms

- LangSmith
  Traces, Evals, Prompt Analytics
- Arize Phoenix
  Model & LLM Observability
- Langfuse
  Traces, Metrics, Observability
- Cloud Observability
  Cloud-native Monitoring & Alerting

From Black Box to Full Visibility

10. AI SECURITY & GUARDRAILS

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AI Security & Guardrails

01 Prompt Injection
02 Indirect Injection
03 Jailbreaking
04 Data Leakage
05 PII Detection
06 Input Guardrails
07 Output Guardrails
08 Tool Permissions
09 Least Privilege
10 Secrets Management
11 Agent Sandboxing
12 MCP Security
13 Rate Limiting
14 AI Red Teaming

Security is a Feature, Not an Afterthought

11. GENAI OPS / AGENTOPS

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GenAI Ops / AgentOps

01 GitHub Actions
02 CI/CD
03 Docker
04 Kubernetes
05 Model Versioning
06 Prompt Versioning
07 Dataset Versioning
08 Experiment Tracking
09 Eval Pipelines
10 Rollbacks
11 Canary Deployment
12 Autoscaling
13 Cost Optimization

MLOps + LLMOps + AgentOps

12. CLOUD AI ENGINEERING

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AWS

- **Amazon Bedrock**
  Foundation Models as a Service
- **SageMaker**
  Build, Train and Deploy ML Models
- **ECS / EKS**
  Run Containerized AI Applications
- **Lambda**
  Serverless AI Workloads
- **CloudWatch**
  Monitoring, Logging and Observability

GCP

- **Vertex AI**
  End-to-End ML Platform
- **Gemini**
  State-of-the-Art Foundation Models
- **Cloud Run**
  Run AI Apps Serverlessly
- **GKE**
  Managed Kubernetes for AI/ML
- **BigQuery**
  Analytics and Data for AI

Azure

- **Azure AI**
  Comprehensive AI Services
- **Azure OpenAI**
  GPT Models in Azure
- **AKS**
  Managed Kubernetes for AI
- **Functions**
  Serverless AI Workloads
- **Azure Monitor**
  Monitoring, Logging and Observability

Deploy AI Anywhere — choose the right cloud, build for scale

13. AI SYSTEM DESIGN

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AI System Design

01 Architecture Design
02 Model Selection
03 RAG vs Fine-Tuning
04 Agent vs Workflow
05 Database Selection
06 Event-Driven Architecture
07 Queues
08 Caching
09 Reliability
10 Fault Tolerance
11 Scalability
12 Security
13 Observability
14 Latency Optimization
15 Cost Engineering

Good Design = Better AI Products